AttriSelect: Efficient Attribute-Driven Data Selection for Targeted Vision Model Repair

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Abstract

Vision models can achieve strong aggregate performance while failing on rare semantic slices that are not defined by class labels alone. Existing attribute-aware debugging and repair pipelines can identify such slices and synthesize targeted repair examples, but they leave an important retraining question under-specified: once generated repair data is available, which original source examples should be retained under a limited retraining budget? We study this question as repair-time source selection. Given a diagnosed failure slice, a source pool with structured attribute annotations, and a generated repair pool, the goal is to select source examples that complement the generated data during retraining. We present AttriSelect, an attribute-driven selector for this setting. AttriSelect scores source examples using class-conditional rarity of repair-relevant attribute values, a penalty for attribute values already over-represented by generated data, and a soft class-balancing correction. We analyze when this score enriches rare target slices relative to attribute-agnostic active-learning selectors, and evaluate it across image classification, pose estimation, and object detection. The results show that explicit attribute metadata provides a useful repair-time selection signal that is distinct from uncertainty, coverage, and gradient-embedding scores (or say traditional active learning methods(TODO)?).

1 Introduction

Computer vision models can perform well on aggregate benchmark metrics while failing on rare semantic subpopulations [4, 6]. A model may recognize people accurately overall but fail on people wearing brown clothes in rainy scenes. These errors are not fully described by the class label: they depend on attribute combinations whose support in the training data may be small [5, 9]. For deployed systems, the practical problem is not only to detect such slices, but also to repair them without repeatedly retraining on all available data.

Attribute-aware debugging methods have made the detection side more interpretable. 3DB [15] constructs structured attribute spaces over model errors, while HiBug [6] uses vision–language models to describe failure slices through shared visual attributes. Related pipelines such as AdaVision [11], DCD [22], and HiBug2 [7] identify underperforming subpopulations and can guide retrieval-style interventions. Generation-based repair provides another route: HiBug [6] and later generation-based methods [25, 23] synthesize targeted examples for discovered slices, with SafeFix [23] using controlled image editing and vision-language filtering to preserve labels and unrelated attributes.

These pipelines address two parts of the repair workflow: which semantic slice is failing, and how to obtain targeted repair examples. A separate question remains: after targeted images have been generated, how should they be combined with the original source data? Retraining on the full source pool can be costly, especially for dense prediction tasks. For example, training Keypoint R-CNN [13]

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for COCO person keypoint estimation [16] takes on the order of eight GPU-days [10]. When new bug slices are discovered iteratively, repeatedly retraining with the full source pool becomes an expensive default.

Subset selection is a natural alternative. Active learning (AL) methods such as uncertainty sampling [21], CoreSet [20], and BADGE [1] select compact training subsets using model uncertainty, feature-space coverage, or gradient embeddings. However, targeted repair differs from generic AL. The selector must preserve useful source coverage while giving enough budget to the diagnosed attribute slice, and it must account for the distribution shift introduced by generated repair images. A generic uncertainty or coverage score may select informative images but still omit rare repair-relevant attribute values; an attribute-greedy policy may do the opposite and over-select values already supplied by generated data.

We therefore present AttriSelect, a repair-time source-data selector that operates after failure-slice discovery and targeted data generation. AttriSelect treats each source image as a structured attribute record rather than only as a point in feature or gradient space. It scores source examples by combining three terms: class-conditional rarity for repair-relevant attributes, a generated-data frequency penalty that discourages duplicating attributes already supplied by synthetic images, and a soft class-deficit correction that limits label or context collapse. The method is intentionally scoped: it does not replace the upstream debugging or generation pipeline, but decides which original examples should accompany the generated repair data during retraining.

Our contributions are:

1. We formulate targeted vision repair as a repair-time source-selection problem: given a diagnosed failure slice, generated repair data, and an annotated source pool, select a compact source subset for retraining.
2. We propose an attribute-driven selector with three separable components: class-conditional rarity, generated-data frequency control, and soft class balancing. This decomposition makes the repair bias explicit and supports direct ablations.
3. We provide a slice-enrichment analysis showing when an attribute-aware score allocates more source budget to rare target slices than attribute-agnostic selectors whose uncertainty or coverage signals are not aligned with the diagnosed slice.
4. We evaluate the selector across image classification, pose estimation, and object detection, comparing against uncertainty, coverage, gradient-embedding, detector-oriented, and random or stratified baselines under matched repair budgets.

2 Related Work

Attribute-Based Debugging and Repair Attribute-based debugging methods aim to expose model failures as human-interpretable semantic slices. AdaVision [11] and DCD [22] identify underperforming subpopulations and rely on retrieval-style interventions, which can require external data sources and manual validation. HiBug [6] uses vision–language models to discover failure slices through shared visual attributes, and HiBug2 [7] further improves the efficiency and coverage of this debugging process. SafeFix [23] shifts from retrieval to controlled generation by editing real training instances and filtering generated images with a vision-language model (VLM) to preserve labels and unrelated attributes. These works focus primarily on finding bug slices or constructing repair examples. We use this line of work as the upstream repair setting and study the remaining selection problem: given original source data and targeted generated data, select a compact retraining subset that preserves useful coverage while emphasizing repair-relevant attributes.

Active Learning Active learning reduces annotation or retraining cost by selecting informative subsets under a budget. Representative approaches include uncertainty sampling [21], coreset selection [20], and hybrid gradient-embedding methods such as BADGE [1]. Recent object-detection and label-budget-robust variants combine these signals, for example PPAL for plug-and-play object detection [24] and Uncertainty Herding for uncertainty-weighted coverage across label budgets [2]. Our setting differs from standard AL because the goal is not generic sample efficiency, but selecting source data that complements a targeted repair set and corrects an attribute-defined failure slice.

3 Background

Attribute-Based Failure Slices Let $f_\theta : \mathcal{X} \to \mathcal{Y}$ denote a vision model, and let $\mathcal{D}_{\text{train}}$, $\mathcal{D}_{\text{val}}$, and $\mathcal{D}_{\text{test}}$ be the training, validation, and test splits. In attribute-aware debugging pipelines such as HiBug [6], each image $x \in \mathcal{X}$ is associated with a collection of semantic attributes $\mathcal{A} = \{a_1, \ldots, a_m\}$, where each attribute function $a_j : \mathcal{X} \to \mathcal{V}_j$ maps an image to a discrete value in a vocabulary $\mathcal{V}_j$. A failure slice is then described by a conjunction of attribute-value assignments,

$$S = \{x \mid a_{j_1}(x) = v_{j_1}, \ldots, a_{j_k}(x) = v_{j_k}\}. \tag{1}$$

Such slices are useful because they turn model failures into human-readable semantic descriptions. We are primarily interested in slices that are both underrepresented in $\mathcal{D}_{\text{train}}$ and underperforming on $\mathcal{D}_{\text{val}}$, i.e., slices whose support is below a rarity threshold ρ and whose validation accuracy is substantially below the overall validation accuracy.

Source Data and Generated Repair Data Once a bug slice has been identified, the repair stage has access to an original source pool $\mathcal{D}_{\text{src}} \subseteq \mathcal{D}_{\text{train}}$ and a generated pool $\mathcal{D}_{\text{gen}}$ of synthetic images for target slices exposed by attribute-aware debugging pipelines such as HiBug and SafeFix [6, 23]. These generated images are constructed to satisfy the target attributes while preserving the original class label. The repair candidate set is

$$\mathcal{D}_{\text{cand}} = \mathcal{D}_{\text{src}} \cup \mathcal{D}_{\text{gen}}. \tag{2}$$

Source data preserves broad training coverage, while generated data adds the missing or rare attribute configurations needed for the discovered repair.

Data Selection for Repair Repair is therefore a budgeted data-curation problem: select a compact retraining set

$$\mathcal{D}_{\text{repair}} = \mathcal{D}_{\text{sel}}^{\text{src}} \cup \mathcal{D}_{\text{sel}}^{\text{gen}}, \tag{3}$$

where $\mathcal{D}_{\text{sel}}^{\text{src}} \subseteq \mathcal{D}_{\text{src}}$ and $\mathcal{D}_{\text{sel}}^{\text{gen}} \subseteq \mathcal{D}_{\text{gen}}$ are chosen under a fixed budget. In the repair runs studied here, the generated repair budget is fixed as 1k images, and the key choice is which source images should accompany it. A useful policy should preserve class coverage, complement the synthetic repair data, and avoid repeatedly selecting attribute values already abundant in $\mathcal{D}_{\text{gen}}$.

4 Methodology

4.1 AttriSelect Framework

AttriSelect is a selection module for a localized repair workflow. The upstream workflow supplies three inputs: a diagnosed attribute slice, a source pool with VLM-assigned attribute values, and a generated repair pool targeting the diagnosed slice. These inputs can come from existing debugging and generation pipelines such as HiBug and SafeFix; AttriSelect does not depend on a specific discovery algorithm or image generator. Its output is a compact source subset $\mathcal{D}_{\text{sel}}^{\text{src}}$ that is combined with the generated repair data for retraining.

This boundary is important for the contribution. Generic active-learning selectors rank source images using model-side quantities such as uncertainty, losses, embeddings, or gradient embeddings. AttriSelect instead ranks them in an explicit attribute-count space tied to the repair slice. The score is decomposed into rarity, generated-data frequency control, and class-deficit correction, so each part corresponds to a concrete repair failure mode and can be removed or tuned independently.

4.2 Attribute-Driven Active Learning

Given $\mathcal{D}_{\text{src}}$ and $\mathcal{D}_{\text{gen}}$ for a discovered bug slice, our goal is to build a compact repair set without expensive retraining on the full source pool. To accelerate repair, a natural strategy is to retrain on a carefully selected subset of the available data rather than on all of $\mathcal{D}_{\text{src}}$. Active learning (AL) offers one principled way to perform this subset selection under a budget. Classical AL includes uncertainty-based strategies that prioritize samples on which the model is least certain, often using scores such as predictive entropy [21], while methods such as CoreSet [20], BADGE [1], PPAL [24], and Uncertainty Herding [2] combine uncertainty, geometric coverage, gradient-space diversity,

or detector-side difficulty signals. However, these methods still operate over image-level model signals and do not explicitly reason about attribute-level semantics or the attribute-distribution shift introduced by targeted synthetic repair data.

This creates two challenges for targeted repair. First, an AL subset can be representative of $\mathcal{D}_{\text{src}}$ in class labels or feature geometry while still containing very few examples from the diagnosed failure slice. If the failure slice is rare, distribution-matching or coverage-based selection allocates only a small fraction of the budget to it; uncertainty sampling can also miss the slice when the model is more uncertain on other images. We formalize this issue in Proposition 1. Second, targeted generation can distort the attribute distribution of the repair pool. For example, if a discovered ImageNet repair attribute is "cloth" or "fabric," a purely attribute-greedy selector may concentrate on classes where that attribute is visually common, such as backpacks or related products, while selecting few examples from classes where the same attribute is rarer, such as guitars with straps or fabric cases. Retraining on such a subset can bias the repaired model toward the classes and contexts that happen to expose the attribute most often, rather than learning a robust attribute-level correction.

Our score is therefore designed for repair rather than generic data summarization. It has three roles:

1. a rarity component R increases the share of source images with rare repair-relevant attributes;
2. a frequency-penalty component P reduces the priority of attribute values already over-supplied by generated data;
3. a class-deficit component C keeps the selected source subset from collapsing onto a small number of classes or pose-complexity bins.

We refer to these three components as R, P, and C throughout the method. Without these components, the repair set can either omit the failure cases, duplicate the synthetic attribute distribution, or overfit to a narrow set of classes.

For each source sample $x_i \in \mathcal{D}_{\text{src}}$ with class label y_i, let $v_{i,a}$ denote the value of attribute a on that image. We restrict attention to a repair-relevant subset $\mathcal{A}_{\text{focus}} \subseteq \mathcal{A}$ of the full attribute set. The first role is handled by the class-conditional rarity term

$$R_{i,a} = \left(\frac{1}{n_a(v_{i,a} \mid y_i) + \varepsilon} \right)^{\alpha} \tag{4}$$

where $n_a(v_{i,a} \mid y_i)$ is the number of source examples in class y_i whose attribute a takes value $v_{i,a}$, $\varepsilon > 0$ is a numerical stability constant, and $\alpha \geq 0$ controls how strongly the selector prefers rare attribute values.

The second role is handled by a frequency penalty based on effective repair-pool counts. We write

$$\hat{n}_a(v) = n_a^{\text{src}}(v) + \tilde{n}_a^{\text{gen}}(v), \tag{5}$$

where $n_a^{\text{src}}(v)$ counts source examples with value v, and $\tilde{n}_a^{\text{gen}}(v)$ denotes the generated-data count used by the selector after any optional scaling of the injected repair data. The penalty is

$$P_{i,a} = -\lambda \cdot \frac{\hat{n}_a(v_{i,a})}{\sum_{v \in \mathcal{V}_a} \hat{n}_a(v)} \tag{6}$$

where $\lambda \geq 0$ controls how strongly the selector penalizes already over-represented attribute values.

The two attribute terms are then summed across focused attributes to obtain the base source score

$$\bar{S}_i = \sum_{a \in \mathcal{A}_{\text{focus}}} \Big(R_{i,a} + P_{i,a} \Big). \tag{7}$$

Removing any of these terms changes the failure mode of the selector. With $\alpha = 0$, the rarity term becomes nearly constant and the selector no longer prefers rare semantic configurations within a class, so existing source failure cases may receive no special priority. With $\lambda = 0$, the source subset can repeat attribute values that were already injected through $\mathcal{D}_{\text{gen}}$, reducing the complementarity between original and generated data and making the repair update more susceptible to synthetic overfitting. With no class-aware re-ranking, the top-scoring images can be concentrated in the classes whose images happen to carry more high-scoring attributes, producing a biased repair subset even when the selected attributes are relevant. The score is therefore not meant to be a neutral representative-sampling score; it is a controlled bias toward repair-relevant semantic coverage.

Soft Class Stratification Attribute scoring alone can over-select classes whose images contain more diverse or more heavily weighted attributes. To counter this effect, we apply a deficit-based class re-ranking step after computing $\bar{S}_i$. Let n_{select} be the source-data budget and let N_{cls} be the number of classes, so that the per-class target is $t = n_{\text{select}}/N_{\text{cls}}$. During greedy selection step k, a candidate from class c receives the adjusted score

$$\tilde{S}_i^{(k)} = \bar{S}_i + C_c^{(k)}, \qquad C_c^{(k)} = w \cdot \frac{t - n_c^{(k)}}{t}, \tag{8}$$

where $C_c^{(k)}$ is a class-deficit correction, $n_c^{(k)}$ is the number of already selected source samples from class c at step k, and $w \geq 0$ controls the strength of class balancing. The correction is positive for classes below their target allocation and negative for classes that have already exceeded it. Larger w pushes the selector closer to stratified sampling; the limiting case $w = \infty$ corresponds to round-robin class quotas. In this way, the method preserves the attribute ranking from Eq. 7 within each class while still preventing a small set of classes from dominating the final repair set.

4.3 Theoretical Analysis

The objective of AttriSelect is not to construct a globally representative coreset of the source data. In targeted repair, a subset is useful only if it contains the rare, underperforming semantic configurations that complement the generated data. We analyze this property as *slice enrichment*. Let $N = |\mathcal{D}_{\text{src}}|$ be the source-pool size and let the source-selection budget be $m = \lfloor rN \rfloor$ for $0 < r < 1$. Let $T \subseteq \mathcal{D}_{\text{src}}$ denote the target failure slice, and write $T_c = T \cap \{x_i : y_i = c\}$ and $N_c = |\{x_i : y_i = c\}|$ for its class-conditional support.

Proposition 1 (Attribute-agnostic AL requires target alignment)**.** *Consider any AL selector that, within class c, chooses B_c source examples using only non-attribute information h_i, such as model uncertainty, logits, feature embeddings, gradient embeddings, losses, detector-side difficulty scores, proxy class-quality weights, coverage gains, or image-level distances, and does not use the focused attribute values $\{v_{i,a} : a \in \mathcal{A}_{\text{focus}}\}$. If the target-slice indicator $\mathbf{1}[x_i \in T_c]$ is conditionally independent of the selector information h_i given class c, with ties and any internal randomness independent of the target indicator, then*

$$\mathbb{E}\big[|S_{\text{AL}} \cap T_c|\big] = B_c \frac{|T_c|}{N_c}. \tag{9}$$

Thus an attribute-agnostic AL method can enrich the target slice only when its non-attribute signal is additionally aligned with that slice.

As a concrete coverage-based instance, let z_i be the representation used by a diversity or coverage selector, such as cluster-based Diversity, CoreSet in feature space, or BADGE in gradient-embedding space. CoreSet-style k-center selection minimizes

$$\max_i \min_{j \in S} d(z_i, z_j), \tag{10}$$

so achieving objective value at most ϵ is equivalent to constructing an ϵ-cover of the source representations. For a finite set A in this representation space, define the empirical covering number

$$\mathcal{N}_\epsilon^{\text{emp}}(A) = \min\big\{|C| : C \subseteq A,\ A \subseteq \cup_{z \in C} B(z, \epsilon)\big\}. \tag{11}$$

If the target slice in class c has empirical covering number

$$q_c = \mathcal{N}_\epsilon^{\text{emp}}\big(\{z_i : x_i \in T_c\}\big), \tag{12}$$

then an ϵ-cover objective can cover the entire target region using only q_c selected target representatives. Consequently, when the non-target region still contains uncovered ϵ-scale cells, a coverage-based selector has an optimal solution with at most q_c selected target examples, even if $|T_c| \gg q_c$.

Proof. For the first claim, conditional independence makes the target indicators exchangeable with respect to any ranking, batch construction, or randomized choice computed from h_i inside class c. The B_c selected positions therefore contain, in expectation, the same target-slice fraction as the class pool itself, namely $|T_c|/N_c$. This argument does not depend on the particular AL heuristic.

Uncertainty sampling, margin sampling, loss-based selection, diversity sampling, CoreSet, BADGE, PPAL-style uncertainty-then-diversity selection, and Uncertainty Herding-style uncertainty coverage all fall under the claim whenever their selection variables omit the current attribute metadata and are not otherwise aligned with the target slice.

For the coverage-based instance, take $h_i = z_i$. By definition of the empirical covering number, there exists a subset $C_T \subseteq \{z_i : x_i \in T_c\}$ with $|C_T| = q_c$ such that every target-slice representation is within distance ϵ of some point in C_T. Selecting the corresponding q_c target examples therefore makes the whole target slice ϵ-covered. Any additional target example only refines coverage inside an already covered target region. If non-target ϵ-scale cells remain uncovered, a coverage or diversity objective can spend the remaining representatives outside T_c without increasing its maximum covering radius. Thus feature-space coverage does not force more than q_c target selections.

The same conclusion follows from the usual volumetric intuition. If a class region has effective dimension d, volume V, and the target subregion has volume V_T, then away from boundary effects an ϵ-cover requires on the order of $V/(v_d\epsilon^d)$ cells for the full class region but only $V_T/(v_d\epsilon^d)$ cells for the target region, where v_d is the volume of the unit d-ball. When the bug slice is semantically important but geometrically compact, $V_T \ll V$, so the number of representatives allocated by a geometry-only objective can be small. The objective allocates capacity to representation volume, not to semantic repair importance. □

The proposition is intentionally stated at the level of selector information rather than at the level of a named baseline. PPAL targets object-detection AL by combining difficulty-calibrated uncertainty with diversity, and Uncertainty Herding targets budget-robust AL by optimizing uncertainty-weighted coverage. These are stronger baselines than plain uncertainty or plain coreset selection, but they still require their uncertainty, difficulty, and coverage signals to be aligned with the discovered attribute slice. If the focused attributes identify a rare failure mode that is not represented in those signals, the expected target-slice allocation remains proportional to the slice prevalence.

We now show why the attribute score assigns strict priority to a rare target slice under explicit data conditions. For any target example $x_i \in T_c$ and non-target example $x_j \notin T_c$ from the same class c, define the focused attributes on which they differ as

$$\mathcal{D}_{ij} = \{a \in \mathcal{A}_{\text{focus}} : v_{i,a} \neq v_{j,a}\}. \tag{13}$$

Let

$$A_{ij} = \sum_{a \in \mathcal{D}_{ij}} \left[\left(\frac{1}{n_a(v_{i,a} \mid c) + \varepsilon} \right)^{\alpha} - \left(\frac{1}{n_a(v_{j,a} \mid c) + \varepsilon} \right)^{\alpha} \right] \tag{14}$$

be the rarity advantage of the target example, and let

$$F_{ij} = \sum_{a \in \mathcal{D}_{ij}} \left[\frac{\hat{n}_a(v_{i,a})}{\sum_{v \in \mathcal{V}_a} \hat{n}_a(v)} - \frac{\hat{n}_a(v_{j,a})}{\sum_{v \in \mathcal{V}_a} \hat{n}_a(v)} \right] \tag{15}$$

be the frequency-penalty disadvantage introduced by source and generated counts. If generated data has flooded the target value, then F_{ij} can be positive; if not, the penalty does not hurt the target example relative to x_j.

Proposition 2 (Rarity-dominates-penalty margin)**.** *Assume $\alpha > 0$ and $A_{ij} > 0$ for every target/non-target pair (x_i, x_j) within class c. Define*

$$\lambda_c^\star = \min_{x_i \in T_c,\, x_j \notin T_c,\, F_{ij} > 0} \frac{A_{ij}}{F_{ij}}, \tag{16}$$

with $\lambda_c^\star = \infty$ if no pair has $F_{ij} > 0$. For any $0 \leq \lambda < \lambda_c^\star$, every target example in T_c has larger base score $\bar{S}_i$ than every non-target example from the same class. In particular, when each target attribute value is rarer than the competing non-target value, i.e., $n_a(v_{i,a} \mid c) < n_a(v_{j,a} \mid c)$ for all $a \in \mathcal{D}_{ij}$, the condition $A_{ij} > 0$ holds automatically.

Proof. For a target/non-target pair (x_i, x_j) in class c, attributes outside $\mathcal{D}_{ij}$ contribute the same score to both examples and cancel. The remaining base-score gap is

$$\bar{S}_i - \bar{S}_j = A_{ij} - \lambda F_{ij}. \tag{17}$$

If $F_{ij} \leq 0$, then $\bar{S}_i - \bar{S}_j \geq A_{ij} > 0$ for any $\lambda \geq 0$, so the frequency penalty does not overturn the rarity advantage. If $F_{ij} > 0$, then $\bar{S}_i - \bar{S}_j > 0$ whenever $\lambda < A_{ij}/F_{ij}$. Taking the minimum over all target/non-target pairs with $F_{ij} > 0$ gives the interval $0 \leq \lambda < \lambda_c^\star$, under which every pair has a strictly positive score gap. Finally, if $n_a(v_{i,a} \mid c) < n_a(v_{j,a} \mid c)$ and $\alpha > 0$, then each rarity term in A_{ij} is positive, so their sum is positive. □

This proposition is the main data-dependent guarantee. It says that if the VLM annotations correctly identify a genuinely rare target slice, then there is a nonempty range of λ values for which rarity dominates the synthetic-data frequency penalty. Even when generated examples make the target value more frequent in the repair candidate pool, the target examples remain ranked above non-target examples as long as λ is below the explicit threshold $\lambda_c^\star$. Conversely, if target and non-target samples are indistinguishable under $\mathcal{A}_{\text{focus}}$, then no attribute-only selector can separate them without additional information.

Proposition 3 (Deterministic rare-slice enrichment)**.** *Assume the rarity-dominates-penalty condition in Proposition 2 holds for class c and the λ used by the selector. If the class-aware re-ranking step allocates B_c source samples to class c, then AttriSelect includes at least*

$$\min\{B_c, |T_c|\} \tag{18}$$

target-slice examples from that class. A random or stratified selector with the same class allocation selects

$$\mathbb{E}\big[|S_{\text{rand}} \cap T_c|\big] = B_c \frac{|T_c|}{N_c} \tag{19}$$

target examples in expectation.

Proof. At a fixed greedy step, the class-aware term adds the same bonus $C_c^{(k)}$ to every remaining candidate from class c. Therefore it can change how many samples are selected from each class, but it cannot reverse the within-class ordering induced by the base attribute score $\bar{S}_i$. By Proposition 2, all target-slice examples in class c appear before all non-target examples in the class queue. Selecting B_c samples from that queue must therefore include all target examples when $|T_c| \leq B_c$, and B_c target examples when $|T_c| > B_c$. This proves the deterministic bound. For random or stratified sampling under the same class allocation, each selected class-c source example has probability $|T_c|/N_c$ of belonging to the target slice, giving the expectation by linearity. □

The enrichment gap is strict whenever $0 < |T_c| < N_c$ and $B_c > 0$, because

$$\min\{B_c, |T_c|\} \geq B_c \frac{|T_c|}{N_c}. \tag{20}$$

Thus AttriSelect provides a semantic coverage guarantee for the repair surrogate, while random, stratified, uncertainty, Diversity, CoreSet, BADGE, PPAL, and Uncertainty Herding require additional alignment between their selection signal and the target slice.

Proposition 4 (Uncertainty can miss an attribute slice)**.** *Consider uncertainty-based selection within each class, with uncertainty score Q_i and class allocation B_c. If the B_c largest uncertainty scores among non-target examples in class c are all larger than every uncertainty score in T_c, then uncertainty selection chooses zero examples from T_c.*

Proof. The claim is deterministic. If at least B_c non-target examples have uncertainty larger than every target example, the top-B_c uncertainty set contains only non-target examples. The same argument applies to any two-stage uncertainty-first method when the first-stage pool contains no target-slice examples: no second-stage diversity or coverage rule can recover target support that is absent from its candidate pool. □

Finally, suppose the repair utility of a selected source subset is

$$U(S) = u_0|S| + (u_T - u_0)|S \cap T|, \qquad u_T > u_0, \tag{21}$$

meaning target-slice samples have larger marginal value for the discovered repair problem. Then the rare-slice enrichment bound directly implies expected utility at least as large as random or stratified selection under the same class allocation, with a strict gap whenever the enrichment gap is strict. This is not a universal proof that the final trained model must dominate every baseline for every ratio; it is a conditional guarantee that the selector spends a fixed budget on more repair-relevant examples when the VLM metadata and count conditions correctly identify the failing slice.

Class-Balance Behavior The enrichment result above is within-class: the class-deficit term changes how many examples are drawn from each class, but it does not reverse the attribute ordering inside a class. The following two results summarize the role of C.

Proposition 5 (Hard-balance limit)**.** *When $w = \infty$ and every active class has enough candidates, the round-robin limit selects class counts that differ by at most one. Within each class, candidates are still ordered by the attribute score $\bar{S}_i$.*

Proof. The round-robin rule visits one active class at a time and removes the highest-scoring remaining candidate from that class. After each visit, the class is moved to the back of the queue. Thus no class can receive a second additional sample before every other non-empty class has received one. The selected counts across non-empty classes therefore differ by at most one, except when a class runs out of candidates. Since each class queue is sorted by $\bar{S}_i$, the within-class attribute order is preserved. □

Proposition 6 (Finite-weight balance pressure)**.** *Assume $0 < w < \infty$. At greedy step k, let $A_c^{(k)}$ be the highest remaining base score in class c, and assume all active class-head scores lie in an interval of width*

$$\Delta = \max_c A_c^{(k)} - \min_c A_c^{(k)}. \tag{22}$$

If class d is selected at step k while another class c is also active, then

$$n_d^{(k)} - n_c^{(k)} \le \frac{t\Delta}{w}. \tag{23}$$

Equivalently, if an active class c trails class d by more than $t\Delta/w$, then class d cannot be selected before class c at that step.

Proof. Since class d is selected, its adjusted score is at least the adjusted score of class c:

$$A_d^{(k)} + w\frac{t - n_d^{(k)}}{t} \ge A_c^{(k)} + w\frac{t - n_c^{(k)}}{t}. \tag{24}$$

Rearranging gives

$$\frac{w}{t}\big(n_d^{(k)} - n_c^{(k)}\big) \le A_d^{(k)} - A_c^{(k)}. \tag{25}$$

Because both class-head scores lie in an interval of width Δ, we have $A_d^{(k)} - A_c^{(k)} \le \Delta$. Therefore $n_d^{(k)} - n_c^{(k)} \le t\Delta/w$. □

This finite-weight result gives the exact sense in which w controls soft balance. When $w = 0$, the bound disappears and the selector reduces to pure attribute ranking. When $0 < w < \infty$, a class with a large deficit receives enough bonus to overcome any bounded attribute-score advantage from an over-selected class. Increasing w tightens the admissible count gap by the factor $1/w$. In the limiting case $w \to \infty$, the admissible gap goes to zero, matching the round-robin behavior in Proposition 5. Thus finite w does not guarantee exact class quotas, but it guarantees that class imbalance can persist only when the over-selected class has a sufficiently larger attribute score at the current step.

5 Experiments

Datasets and Tasks We evaluate AttriSelect on three vision tasks: image classification, pose estimation, and object detection. In all settings, the goal is to test whether attribute-aware source-data selection can choose a subset that improves the model performance and fixes bugs.

For image classification, we follow SafeFix [23]: a ResNet-18 classifier on an ImageNet subset [8] with 10 classes, 1,300 images per class, and an 8:1:1 train/validation/test split, denoted ImageNet10. We also evaluate CelebA [17] lipstick-wearing classification with a ResNet-18-based binary classifier and 80,000/10,000/10,000 train/validation/test images. For pose estimation, we follow HiBug2 [7] on COCO person keypoints [16], using Keypoint R-CNN [13] with a ResNet-50-FPN backbone, 20,000 source training images, and 1,173-image held-out validation and test splits, evaluated by COCO keypoint AP. For object detection, we use KITTI [12] Car and Pedestrian instances converted to YOLO format with 5,000/1,240/1,241 train/validation/test images, and train YOLOv8n.

Table 1: ImageNet10 ControlNet repair results. Metric: FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	3.40	3.47	2.87	3.40	5.31	5.59	3.96	**6.93**
0.3	4.78	6.93	4.21	6.93	8.84	10.47	7.75	**11.28**
0.5	7.32	9.66	5.59	8.03	10.47	8.38	8.84	**11.85**
0.7	8.84	6.12	6.40	7.75	10.75	10.47	8.84	**12.10**
0.9	10.22	10.47	10.22	6.12	9.37	11.57	11.85	**13.76**
1.0	8.03	8.03	8.03	8.03	8.03	8.03	8.03	8.03

Table 2: CelebA lipstick-wearing classification repair results. Metric: FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	4.88	2.44	7.10	-6.04	3.18	-1.06	5.94	**8.48**
0.3	6.04	5.30	8.70	7.95	1.91	3.50	4.03	**8.91**
0.5	6.04	7.95	7.64	8.17	2.86	2.23	9.97	**13.79**
0.7	2.86	8.27	10.18	6.57	8.06	5.09	6.04	**12.09**
0.9	3.18	4.56	8.27	6.04	6.04	4.56	6.57	**14.74**
1.0	6.04	6.04	6.04	6.04	6.04	6.04	6.04	6.04

Metrics Let M denote the task metric: test accuracy for image classification, COCO keypoint AP for pose estimation, and mAP50–95 for object detection. We focus on targeted improvements for failure cases, which are critical for fairness and safety, while also tracking overall task performance. Following prior work [14, 23], we report the Fix Rate (FR),

$$\mathrm{FR}(\%) = 100 \cdot \frac{M_{\mathrm{after}} - M_{\mathrm{before}}}{1 - M_{\mathrm{before}}}, \tag{26}$$

where M_{before} and M_{after} are the test metrics before and after repair. For accuracy, FR is the percentage of the original error gap that is corrected; for AP and mAP, it is the normalized improvement toward the maximum score, reported in percent.

Baselines We compare AttriSelect with seven generic source-selection baselines. RANDOM samples source images uniformly at random. STRATIFIED preserves class-level proportions during sampling [18]. ENTROPY selects examples with the largest predictive entropy [21]. CORESET uses greedy coverage in embedding space [20]. BADGE selects gradient-embedding batches that combine uncertainty and diversity [1]. PPAL is a plug-and-play active-learning baseline for object detection [24]. UHERDING denotes Uncertainty Herding, which combines uncertainty with coverage across label budgets [2].

Implementation Details We use vision-language models (VLMs) in two separate roles. First, a VLM proposes the semantic attribute vocabulary $\mathcal{A}$ for a task. For example, in KITTI object detection, GPT-5 [19] is used to define candidate visual attributes such as color, weather, scale, viewpoint, occlusion, truncation, and motion state. Second, a VLM assigns attribute values to the source images. In our KITTI implementation, we use Qwen3-VL [3] for this assignment step. The selector then uses only the repair-relevant subset $\mathcal{A}_{\mathrm{focus}} \subseteq \mathcal{A}$ in Eq. 7; for the red-cloudy KITTI repair, this subset contains color and weather. This separation lets the full attribute space describe the dataset broadly, while the attribute-selection score focuses on the attributes tied to the discovered repair slice.

5.1 Main Results

Tables 1–4 compare AttriSelect with generic source-selection baselines using Fix Rate. The ratio 1.0 rows use the full original source set, so subset selection is inactive and the repeated entries are the corresponding full-data reference. The original task metrics are reported in Appendix A.

5.2 Computational Efficiency

AttriSelect is also cheaper than generic active-learning selectors. Its selection cost is dominated by counting VLM-assigned attribute values and sorting candidates, while uncertainty, coverage, and

Table 3: COCO brown-rainy pose repair results. Metric: FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	3.56	5.08	5.61	-0.61	2.03	2.71	1.17	**6.86**
0.3	3.37	8.32	7.64	8.64	6.12	4.32	6.42	**9.86**
0.5	7.49	8.00	9.27	6.47	7.42	6.12	6.83	**10.13**
0.7	6.08	9.69	9.54	9.10	8.64	6.17	7.83	**10.18**
0.9	4.61	9.49	6.54	7.32	7.59	7.05	9.74	**11.05**
1.0	8.71	8.71	8.71	8.71	8.71	8.71	8.71	8.71

Table 4: KITTI red-cloudy object-detection repair results. Metric: FR (%).

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	-9.65	-7.27	-7.52	-8.89	-6.95	-9.24	-8.32	**-5.81**
0.3	-0.48	-1.23	0.23	-0.32	-0.05	-1.21	-1.33	**0.46**
0.5	2.13	1.23	3.20	3.04	2.29	1.12	2.17	**3.36**
0.7	3.20	3.06	3.34	3.34	2.08	3.25	**3.57**	3.52
0.9	4.37	4.66	4.66	3.54	4.21	4.85	3.89	**5.12**
1.0	5.30	5.30	5.30	5.30	5.30	5.30	5.30	5.30

gradient-embedding baselines require model scoring, embedding comparisons, or iterative diversity updates. Table 5 reports source-subset selection time on COCO and CelebA for the three ratios used in the main small-to-medium budget comparison. The source pools contain 5,000 COCO training images and 80,000 CelebA training images. Random and stratified sampling have negligible algorithmic selection cost after excluding data loading and export overhead; we keep them in the table as zero-cost references.

Across these settings, AttriSelect selects the source subset in less than two seconds, whereas the fastest non-attribute active-learning baseline takes tens of seconds and the slower coverage or gradient-embedding methods can take minutes to hours. The efficiency gain is therefore not only relative to the most expensive baselines: even against the fastest nontrivial baseline in each row, AttriSelect is at least $36\times$ faster on CelebA and more than $230\times$ faster on COCO.

5.3 Attribute-Slice Accuracy Deltas

Figure 1 uses ImageNet10 accuracy deltas as a slice-level example of what the source selector changes. The generated repair data targets images with pink color and fabric texture; all methods use the same generated pool and differ only in which ratio-0.5 source subset is retained for retraining. The attribute-aware source subset selected by AttriSelect yields substantially larger gains than the active-learning baselines on the targeted attribute groups. The exact repair values, pink and fabric, receive the largest improvements. Non-targeted attributes show smaller and less uniform changes, with occasional decreases, but AttriSelect still tends to give slightly larger gains across these rows because the class-balance and frequency terms preserve broader source coverage while prioritizing the repair attributes. For readability, this main-text analysis shows three ImageNet10 attribute groups: color, background, and texture. A larger version with all annotated COCO pose-estimation attribute groups is provided in Appendix B.

5.4 Ablation Study

We ablate the three score components on ImageNet10 ControlNet repair to test whether the gains come from a single term or from their combination. Table 6 reports test accuracy for all component subsets. The full $R + P + C$ setting combines rarity R, frequency penalty P, and class-deficit correction C.

The full $R + P + C$ setting is the most reliable pattern, giving the best result at every ratio. The leave-one-component-out drops from the full setting are nearly identical: removing R drops test accuracy by 0.00942 on average, removing P by 0.00940, and removing C by 0.00940. Thus the ablation supports a complementarity claim rather than a single dominant-component claim on ImageNet10. Single-component variants are weaker on average (R: 0.74198, P: 0.74038, C: 0.73814) than the

Table 5: Selection-stage wall-clock time. Units are seconds (s), minutes (m), and hours (h). Speedup is measured against the fastest non-attribute active-learning baseline in the same row.

Dataset	Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours / speedup**
COCO	0.1	0s	0s	863s	5m	21m	86s	67s	**0.29s / 231×**
COCO	0.3	0s	0s	865s	15m	1h	93s	85s	**0.30s / 283×**
COCO	0.5	0s	0s	865s	25m	2h	94s	90s	**0.32s / 281×**
CelebA	0.1	0s	0s	49s	45m	2h	63s	46s	**0.42s / 110×**
CelebA	0.3	0s	0s	51s	3h	5h	65s	51s	**0.51s / 100×**
CelebA	0.5	0s	0s	49s	5h	9h	61s	61s	**1.37s / 36×**

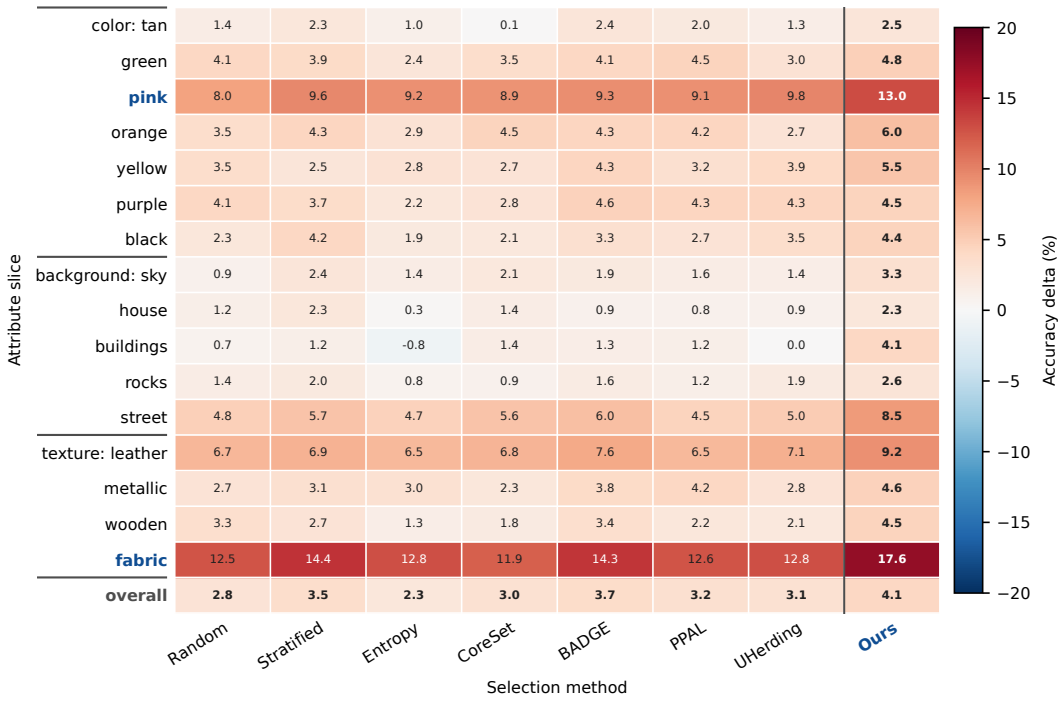


Figure 1: ImageNet10 ratio-0.5 attribute-slice accuracy deltas. The upstream repair pipeline generates examples targeting pink color and fabric texture, and each method retrains with the same generated data plus a ratio-0.5 selected source subset. AttriSelect gives larger gains on the targeted attribute families, with the strongest gains on the exact repair values.

full setting (0.74892), showing that each component alone is insufficient. A COCO Brown-Rainy component ablation with a broadly similar trend is reported in Appendix D.

6 Conclusion

We studied targeted vision model repair as a repair-time source-selection problem. In this setting, a failure slice has already been diagnosed and targeted repair images have already been generated; the remaining question is which original source examples should accompany those generated images during retraining. AttriSelect addresses this question with an attribute-driven score that combines class-conditional rarity, generated-data frequency control, and soft class balancing. The analysis shows when this score enriches rare target slices relative to attribute-agnostic selectors, and the experiments across classification, pose estimation, and detection indicate that structured attribute metadata provides a useful signal beyond generic uncertainty, coverage, and gradient-embedding scores. This makes AttriSelect a focused component for localized repair pipelines rather than a replacement for upstream bug discovery or image generation.

Table 6: ImageNet10 ControlNet attribute-score component ablation. Metric: test accuracy.

Ratio	none	R	P	C	$P+C$	$R+C$	$R+P$	$R+P+C$
0.1	0.7269	0.7300	0.7262	0.7277	0.7359	0.7308	0.7315	**0.7369**
0.3	0.7308	0.7415	0.7392	0.7392	0.7308	0.7369	0.7315	**0.7492**
0.5	0.7400	0.7415	0.7408	0.7346	0.7369	0.7415	0.7446	**0.7508**
0.7	0.7423	0.7469	0.7431	0.7415	0.7431	0.7469	0.7477	**0.7515**
0.9	0.7462	0.7500	0.7526	0.7477	0.7508	0.7415	0.7423	**0.7562**

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Appendix

A Main-Result Metric Tables

This section reports the original task metrics corresponding to the Fix Rate tables in the main text. The metrics are test accuracy for ImageNet10 and CelebA, COCO keypoint AP for pose estimation, and test mAP50–95 for KITTI object detection.

Table 7: ImageNet10 ControlNet repair results. Metric: test accuracy.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	0.7269	0.7271	0.7254	0.7269	0.7323	0.7331	0.7285	**0.7369**
0.3	0.7308	0.7369	0.7292	0.7369	0.7423	0.7469	0.7392	**0.7492**
0.5	0.7380	0.7446	0.7331	0.7400	0.7469	0.7410	0.7423	**0.7508**
0.7	0.7423	0.7346	0.7354	0.7392	0.7477	0.7469	0.7423	**0.7515**
0.9	0.7462	0.7469	0.7462	0.7346	0.7438	0.7500	0.7508	**0.7562**
1.0	0.7400	0.7400	0.7400	0.7400	0.7400	0.7400	0.7400	0.7400

Table 8: CelebA lipstick-wearing classification repair results. Metric: test accuracy.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	0.9103	0.9080	0.9124	0.9000	0.9087	0.9047	0.9113	**0.9137**
0.3	0.9114	0.9107	0.9139	0.9132	0.9075	0.9090	0.9095	**0.9141**
0.5	0.9114	0.9132	0.9129	0.9134	0.9084	0.9078	0.9151	**0.9187**
0.7	0.9084	0.9135	0.9153	0.9119	0.9133	0.9105	0.9114	**0.9171**
0.9	0.9087	0.9100	0.9135	0.9114	0.9114	0.9100	0.9119	**0.9196**
1.0	0.9114	0.9114	0.9114	0.9114	0.9114	0.9114	0.9114	0.9114

Table 9: COCO brown-rainy pose repair results. Metric: COCO keypoint AP.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	0.6048	0.6110	0.6132	0.5877	0.5985	0.6013	0.5950	**0.6183**
0.3	0.6040	0.6243	0.6215	0.6256	0.6153	0.6079	0.6165	**0.6306**
0.5	0.6209	0.6230	0.6282	0.6167	0.6206	0.6153	0.6182	**0.6317**
0.7	0.6151	0.6299	0.6293	0.6275	0.6256	0.6155	0.6223	**0.6319**
0.9	0.6091	0.6291	0.6170	0.6202	0.6213	0.6191	0.6301	**0.6355**
1.0	0.6259	0.6259	0.6259	0.6259	0.6259	0.6259	0.6259	0.6259

B Attribute-Slice Heatmaps

Figure 2 reports the corresponding ratio-0.5 COCO pose-estimation slice analysis over the full annotated attribute vocabulary. The targeted repair values are clothes color brown and rainy weather. As in the ImageNet10 example, the largest gains appear on the targeted values, while the remaining attributes show smaller changes.

C Hyperparameter Details

Table 11 reports the AttriSelect hyperparameters used in the main experiments. The parameters are the rarity exponent α, frequency-penalty weight λ, and class-balancing weight w, matching the notation in Eq. 8.

The larger frequency-penalty weight used for KITTI is consistent with the relative size of the generated repair set. All four settings add 1k generated images, but KITTI has only 5,000 source training images, compared with 10,400 for ImageNet10, 20,000 for COCO, and 80,000 for CelebA. The generated images therefore form a larger fraction of the KITTI repair candidate pool, making a stronger penalty on already-supplied attribute values useful for preserving source-data complementarity. This dataset-dependent shift supports the design of the score: the frequency-penalty term adapts the selector to the amount of generated repair data relative to the available source pool.

Table 12 gives the KITTI sensitivity across source-selection ratios. Across several AttriSelect parameter settings, the test mAP50–95 changes only modestly, suggesting that the selector is not brittle to small changes in α, λ, and w once the generated-data frequency penalty is present.

Table 13 shows the same pattern on COCO Brown-Rainy pose repair. The selected main setting remains strongest for most ratios, while nearby AttriSelect parameter settings stay within a narrow AP band.

D COCO Brown-Rainy Attribute-Score Component Ablation

Table 14 reports the corresponding component ablation for COCO Brown-Rainy pose repair. The full $R + P + C$ setting is strongest at four of the five ratios and remains close to the best result at the remaining ratio, suggesting that the three terms are complementary in this setting as well.

The leave-one-component-out comparisons show that removing any term reduces average AP relative to the full score, with average drops ranging from 0.00708 to 0.01428. Single-component variants are also less reliable than the full score across ratios. Overall, the COCO ablation supports the same

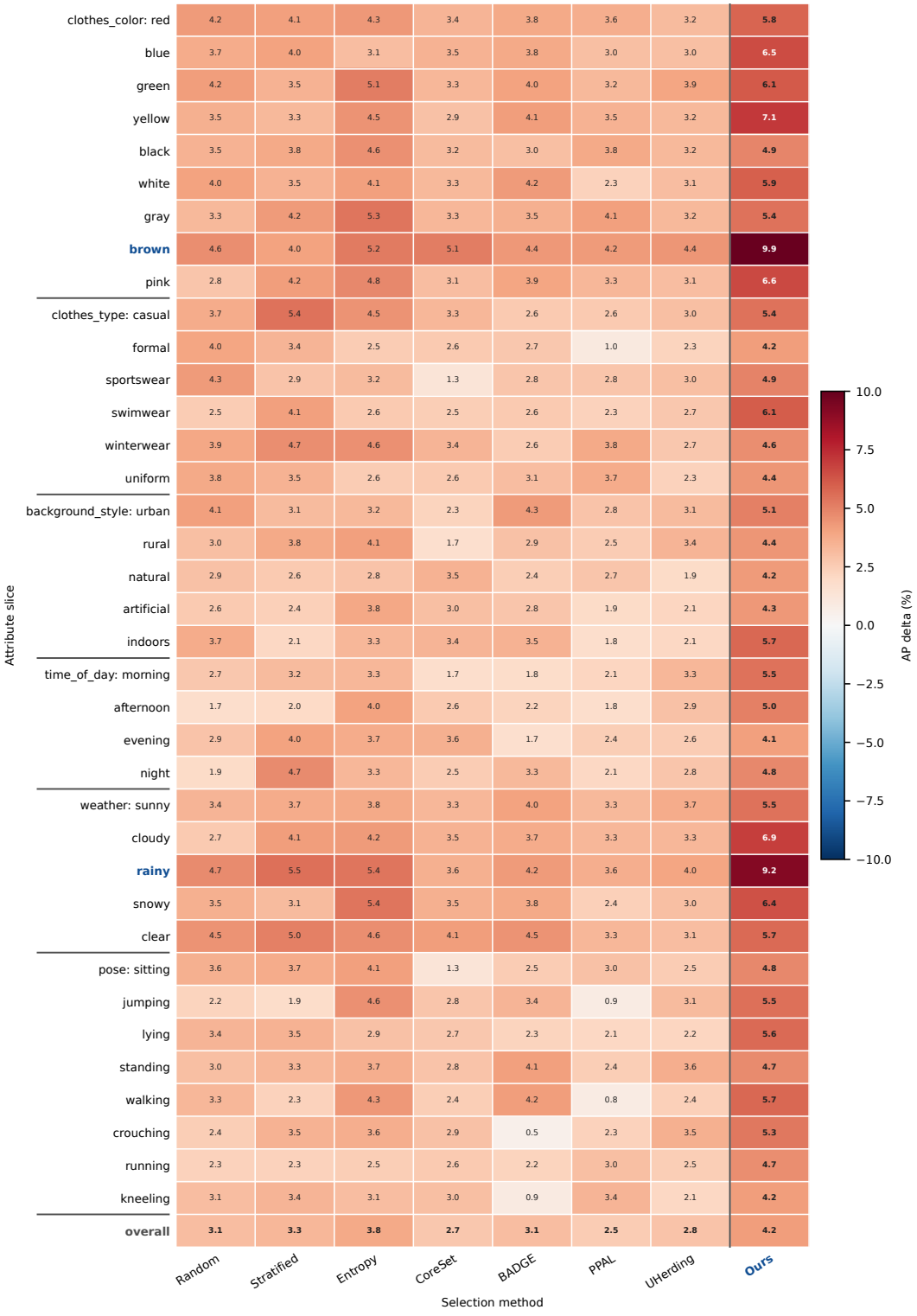


Figure 2: COCO brown-rainy ratio-0.5 attribute-slice AP deltas. The figure expands the slice-level analysis to all annotated COCO pose-estimation attribute groups. The targeted repair values are clothes color brown and rainy weather.

Table 10: KITTI red-cloudy object-detection repair results. Metric: test mAP50–95.

Ratio	Random	Stratified	Entropy	CoreSet	BADGE	UHerding	PPAL	**Ours**
0.1	0.5204	0.5308	0.5297	0.5237	0.5322	0.5222	0.5262	**0.5372**
0.3	0.5605	0.5572	0.5636	0.5612	0.5624	0.5573	0.5568	**0.5646**
0.5	0.5719	0.5680	0.5766	0.5759	0.5726	0.5675	0.5721	**0.5773**
0.7	0.5766	0.5760	0.5772	0.5772	0.5717	0.5768	**0.5782**	0.5780
0.9	0.5817	0.5830	0.5830	0.5781	0.5810	0.5838	0.5796	**0.5850**
1.0	0.5858	0.5858	0.5858	0.5858	0.5858	0.5858	0.5858	0.5858

Table 11: AttriSelect hyperparameters used in the main experiments.

Dataset	α	λ	w
ImageNet10	0.03	0.5	15
CelebA	0.03	0.5	15
COCO	0.03	0.5	15
KITTI	0.01	5.0	15

conclusion as ImageNet10: the best repair behavior comes from combining rarity, generated-data frequency control, and class balancing rather than relying on any one term alone.

Table 12: KITTI AttriSelect sensitivity. Metric: test mAP50–95.

α	λ	w	0.1	0.3	0.5	0.7	0.9
0.01	5.0	15	**0.5372**	**0.5646**	**0.5773**	0.5780	**0.5850**
0.03	0.5	15	0.5358	0.5637	0.5764	0.5774	0.5842
0.03	5.0	8	0.5366	0.5641	0.5769	**0.5784**	0.5847
0.10	5.0	15	0.5349	0.5640	0.5768	0.5776	0.5839
0.80	10.0	4	0.5337	0.5629	0.5756	0.5769	0.5828

Table 13: COCO AttriSelect sensitivity. Metric: COCO keypoint AP.

α	λ	w	0.1	0.3	0.5	0.7	0.9
0.03	0.5	15	**0.6183**	**0.6306**	**0.6317**	0.6319	**0.6355**
0.01	0.5	15	0.6152	0.6298	0.6309	0.6308	0.6337
0.03	0.5	8	0.6165	0.6289	0.6305	**0.6323**	0.6344
0.03	0.5	0	0.6148	0.6276	0.6292	0.6306	0.6328
0.00	0.0	15	0.6042	0.6288	0.6203	0.6313	0.6268

Table 14: COCO Brown-Rainy attribute-score component ablation. Metric: COCO keypoint AP.

Ratio	none	R	P	C	$P+C$	$R+C$	$R+P$	$R+P+C$
0.1	0.6048	0.6099	0.5983	0.6067	0.6066	0.6149	0.5954	**0.6183**
0.3	0.6040	0.6134	0.6038	0.6136	0.6142	0.6196	0.6136	**0.6306**
0.5	0.6209	0.6074	0.6113	0.6149	0.6295	0.6246	0.6155	**0.6317**
0.7	0.6151	0.6222	0.6231	**0.6319**	0.6317	0.6302	0.6282	0.6311
0.9	0.6091	0.6175	0.6174	0.6270	0.6276	0.6225	0.6231	**0.6355**

E No-Generation Corner Case

This section discusses the corner case in which no generated repair images are available, i.e., $\mathcal{D}_{\text{gen}} = \emptyset$. This setting removes the main mechanism for adding missing semantic support, but AttriSelect still has a useful source-selection interpretation. Since $\tilde{n}_a^{\text{gen}}(v) = 0$, the effective count becomes

$$\hat{n}_a(v) = n_a^{\text{src}}(v), \tag{27}$$

and the base score reduces to

$$\bar{S}_i = \sum_{a \in \mathcal{A}_{\text{focus}}} \left[\left(\frac{1}{n_a(v_{i,a} \mid y_i) + \varepsilon} \right)^{\alpha} - \lambda \frac{n_a^{\text{src}}(v_{i,a})}{\sum_{v \in \mathcal{V}_a} n_a^{\text{src}}(v)} \right]. \tag{28}$$

Thus, when $\mathcal{D}_{\text{gen}} = \emptyset$, the frequency penalty is no longer correcting a synthetic flood. It becomes a source-distribution head-value penalty: values that are already common in the source pool receive lower priority. The rarity term still prefers values that are rare within a class, and the class-balance term still controls label-level collapse.

Proposition 7 (Source-only rare-slice enrichment)**.** *Let $\mathcal{D}_{\text{gen}} = \emptyset$ and let $T \subseteq \mathcal{D}_{\text{src}}$ be a target failure slice with nonzero source support. Suppose the rarity-dominates-penalty condition in Proposition 2 holds for T under the source-only score above, with $\hat{n}_a(v) = n_a^{\text{src}}(v)$. If the class-aware re-ranking step selects B_c samples from class c, then AttriSelect returns at least*

$$\sum_c \min\{B_c, |T_c|\} \tag{29}$$

source examples from T. A random selector with the same per-class allocation selects

$$\mathbb{E}\big[|S_{\text{rand}} \cap T|\big] = \sum_c B_c \frac{|T_c|}{N_c} \tag{30}$$

target-slice examples in expectation.

Proof. The proof is identical to the rare-slice enrichment proof in the main text, because that argument uses only the within-class ordering induced by $\bar{S}_i$. When $\mathcal{D}_{\text{gen}} = \emptyset$, the score changes only through the replacement $\hat{n}_a(v) = n_a^{\text{src}}(v)$. If the source-only score satisfies the rarity-dominates-penalty condition, then all target-slice samples appear before non-target samples in each class queue. Selecting B_c examples from class c therefore includes $\min\{B_c, |T_c|\}$ target examples. The random expectation follows by linearity under sampling from each class with allocation B_c. □

This proposition shows that generation is not required for the selector to improve target-slice coverage when the source pool already contains target examples. In this regime, the method becomes semantic active subset selection over the source pool. It can increase the number of existing rare failure cases in the retraining subset, but it cannot create support for a target slice that is absent from the source pool.

Proposition 8 (Impossibility for absent source slices)**.** *Let $\mathcal{D}_{\text{gen}} = \emptyset$ and suppose $T \cap \mathcal{D}_{\text{src}} = \emptyset$. Then every source-only selector $S \subseteq \mathcal{D}_{\text{src}}$ satisfies*

$$|S \cap T| = 0. \tag{31}$$

Consequently, no active-selection rule over $\mathcal{D}_{\text{src}}$ alone can provide target-slice training support.

Proof. Since $S \subseteq \mathcal{D}_{\text{src}}$ and $T \cap \mathcal{D}_{\text{src}} = \emptyset$, we have $S \cap T \subseteq \mathcal{D}_{\text{src}} \cap T = \emptyset$. Hence $|S \cap T| = 0$ for every possible selector. □

The no-generation case therefore separates two roles of AttriSelect. If T already appears in the source pool, attribute selection can enrich those existing target examples under the same rarity-dominates-penalty condition as the main result. If T is absent from the source pool, selection alone is insufficient; repair requires either new data collection or generated images that add target-slice support. This also clarifies the role of λ: with no generated images, a large frequency penalty may suppress globally common values even when they are rare within a specific class. In source-only repair, λ should therefore be interpreted as a head-value suppression weight and may need to be smaller than in settings with heavy generated-data injection.

F Selection t-SNE Visualizations

Before measuring selected-set overlap directly, we first visualize the selected subsets in a shared low-dimensional embedding space. These plots are qualitative: the t-SNE coordinates are computed from lightweight image proxy embeddings and are used only for visualization, not for the selection-overlap metric below. Gray points denote source candidates, and colored points denote the subset selected by each method.

The ratio-0.10 plots in Figures 3–6 cover all four evaluated datasets. At this low selection ratio, each method selects a relatively small subset, so differences in coverage are visually readable. The available ratio-0.05 plots in Figures 8 and 9 provide a more stringent small-budget view for the datasets where this auxiliary diagnostic was run. CelebA and COCO are omitted at ratio 0.05 because the corresponding small-budget diagnostic was not included for those datasets. Figures 7 and 10 show compact KITTI overlays for a subset of methods, making the separation between AttriSelect and representative active-learning baselines easier to inspect.

The visual pattern should be interpreted in light of the selection score in Section 4.3. AttriSelect is not designed to be a uniformly spread coreset in the t-SNE plane. The rarity term $R_{i,a}$ intentionally gives higher priority to source examples whose focused attribute values are rare within their class, so the selected points can concentrate in regions that contain rare repair-relevant appearances. The penalty term $P_{i,a}$ reduces the priority of attribute values already supplied by the generated pool, so the selected source points are encouraged to complement the synthetic repair data instead of repeating it. Finally, the class-deficit correction $C_c^{(k)}$ pulls the greedy selection away from a single high-scoring class or local region. Thus the expected signature is controlled non-uniformity: AttriSelect may put more mass on visually or semantically unusual regions than random, uncertainty, or pure coverage baselines, while still retaining cross-class coverage through the balance term.

The overlay plots illustrate this qualitative behavior on KITTI. The AttriSelect points do not simply trace the same support as the representative active-learning baselines; they are shifted toward a different subset of the proxy embedding space. This is consistent with the score's purpose: select source images that are representative enough to support retraining, but biased toward rare and repair-relevant attribute configurations. Because t-SNE is only a two-dimensional proxy visualization, this observation is not a standalone performance claim. It motivates the direct selected-set overlap analysis in Section G, where the same conclusion is tested through Jaccard overlap between selected subsets.

These visualizations should not be interpreted as proof that low overlap improves downstream repair. They instead provide a sanity check that AttriSelect is not visibly collapsing onto the same region as the standard active-learning selectors. The quantitative version of this diagnostic is the Jaccard analysis in Section G.

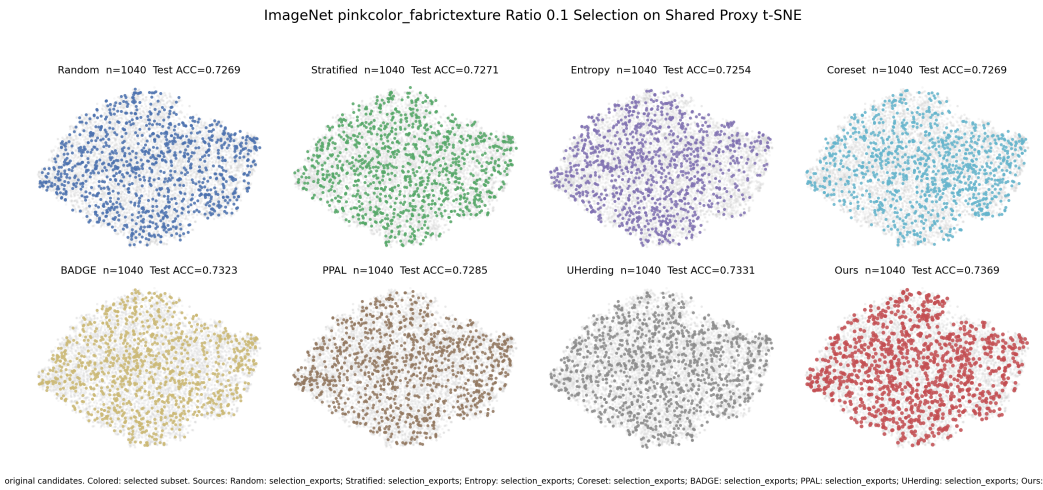


Figure 3: ImageNet ratio-0.10 selected subsets visualized on a shared source-image t-SNE embedding. Gray points show source candidates; colored points show the subset selected by each method.

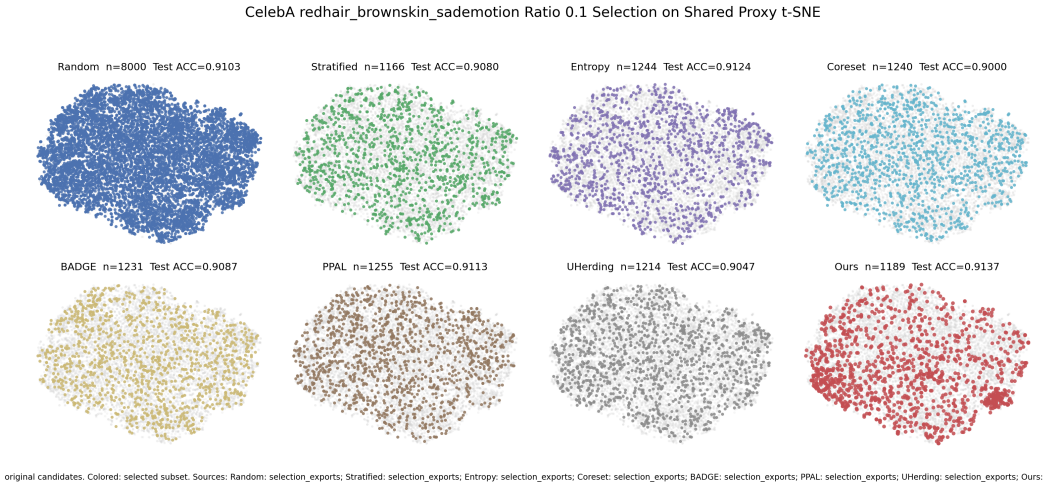


Figure 4: CelebA ratio-0.10 selected subsets visualized on a shared source-image t-SNE embedding.

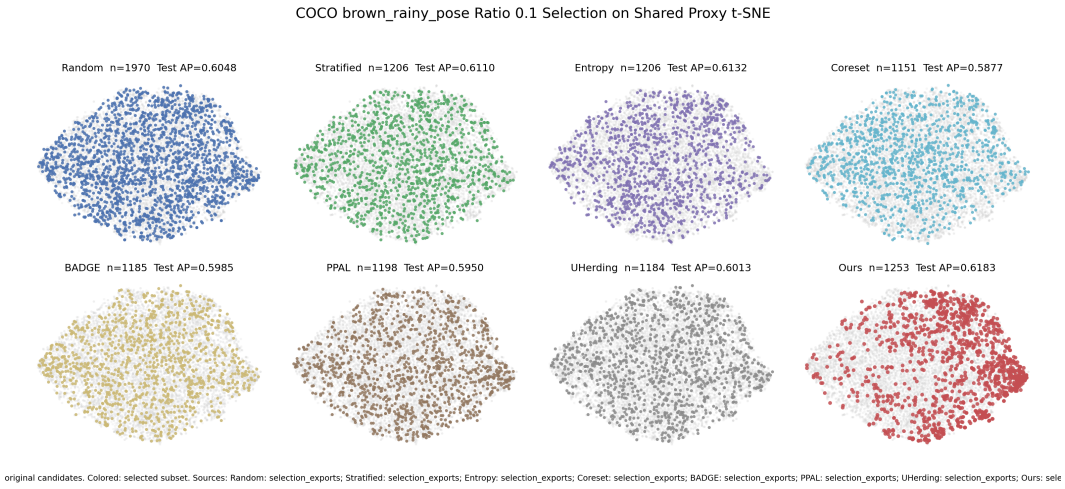


Figure 5: COCO Brown-Rainy ratio-0.10 selected subsets visualized on a shared source-image t-SNE embedding.

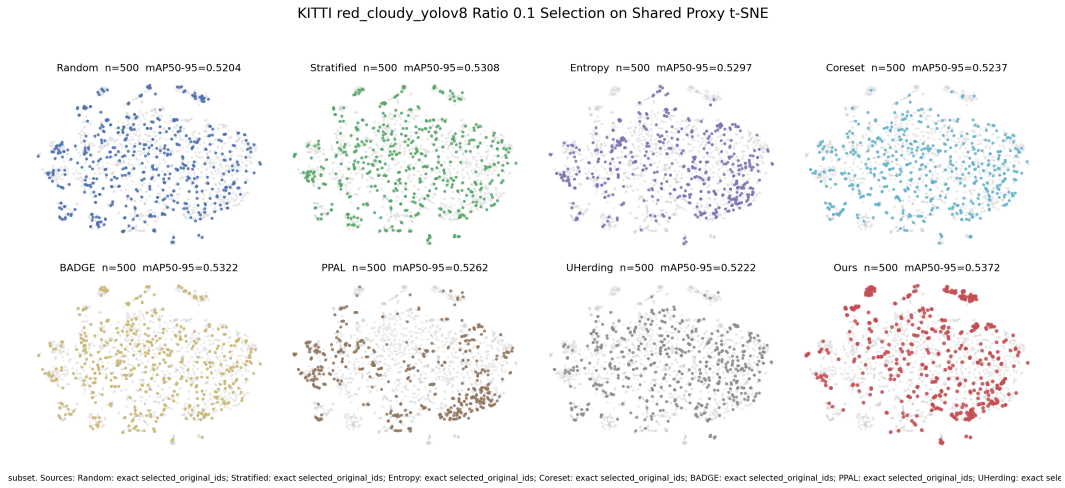


Figure 6: KITTI Red-Cloudy ratio-0.10 selected subsets visualized on a shared source-image t-SNE embedding.

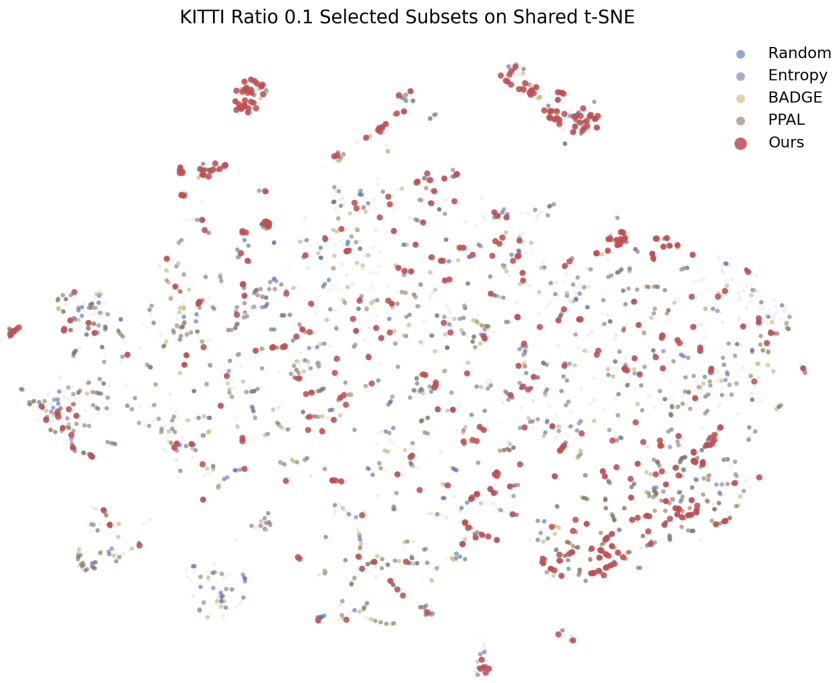


Figure 7: KITTI Red-Cloudy ratio-0.10 overlay view for selected active-learning baselines and AttriSelect.

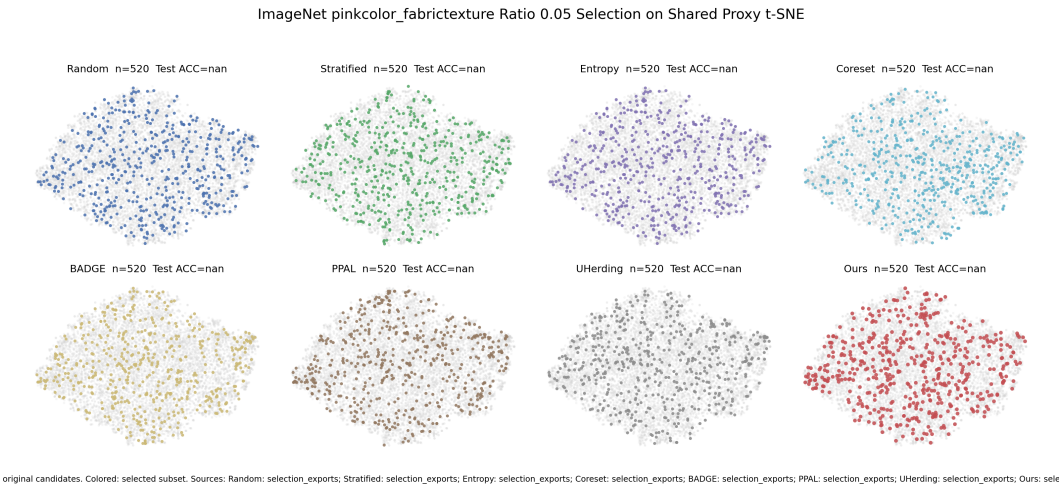


Figure 8: ImageNet ratio-0.05 selected subsets visualized on a shared source-image t-SNE embedding.

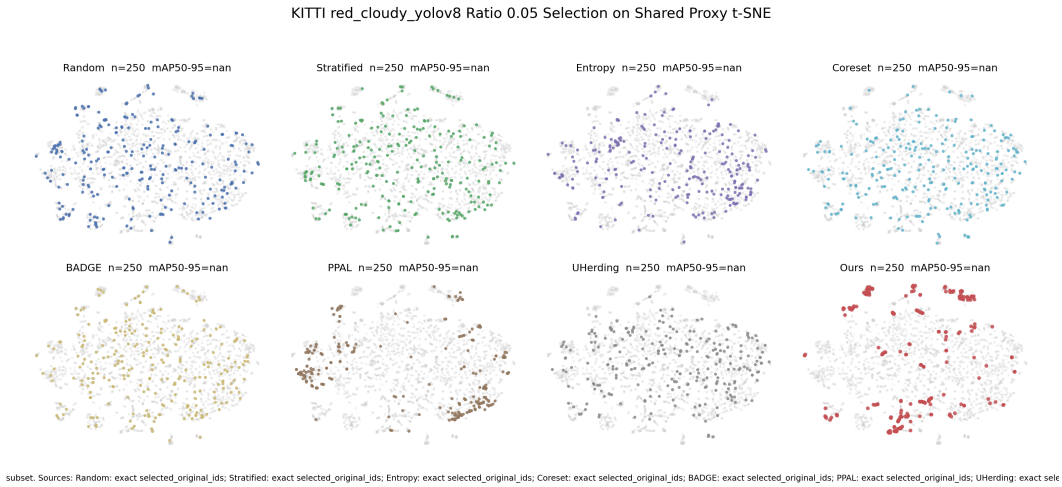


Figure 9: KITTI Red-Cloudy ratio-0.05 selected subsets visualized on a shared source-image t-SNE embedding.

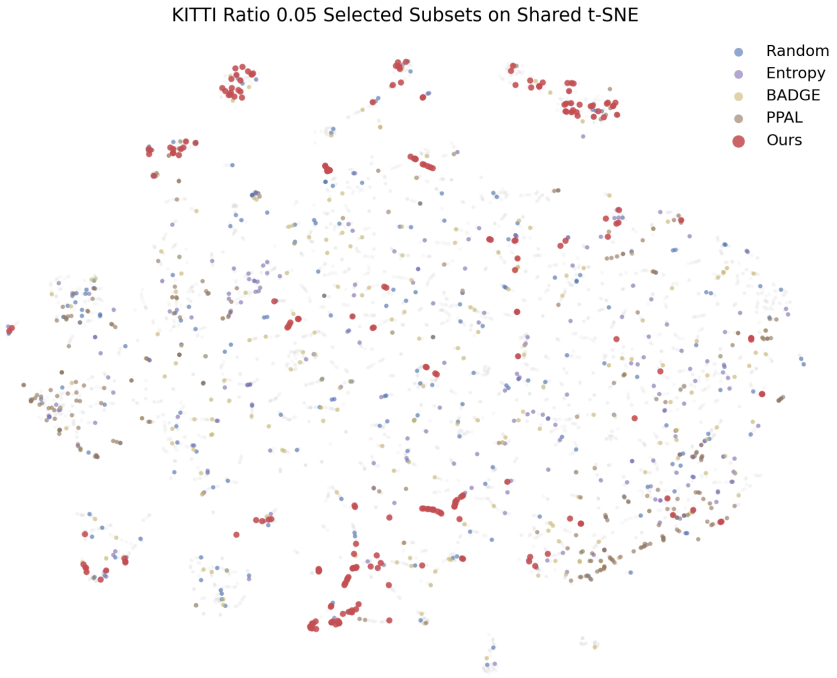


Figure 10: KITTI Red-Cloudy ratio-0.05 overlay view for selected active-learning baselines and AttriSelect.

G Selection Overlap as an Attribute-Information Diagnostic

This section formalizes what the selection-overlap plots test. Standard active learning selectors rank or cover examples using information such as logits, losses, uncertainty scores, feature embeddings, gradient embeddings, image distances, and sometimes class labels. Let H_i denote all such non-attribute information for source example x_i. Our selector additionally observes structured attribute metadata

$$A_i = \big(a(x_i) : a \in \mathcal{A}_{\text{focus}}\big). \tag{32}$$

Thus the comparison is not only between two scoring formulas. It is a comparison between selectors that are measurable with respect to different information fields: generic AL is attribute-agnostic when it is a function of H_i, while the proposed selector is a function of both H_i and A_i through rarity, generated-data frequency, and class-aware re-ranking.

The overlap plots therefore answer a specific question: is AttriSelect merely reproducing the same subset chosen by uncertainty, CoreSet, BADGE, PPAL, Uncertainty Herding, or random selection? Low overlap by itself is not a performance guarantee. However, when low overlap is paired with the slice-enrichment result in Proposition 2 and with improved downstream repair, it supports a stronger interpretation: the gain comes from a semantic selection direction that is not already captured by generic AL signals.

Proposition 9 (Random-overlap scale for attribute-agnostic selectors)**.** *Let the source pool be partitioned into groups $\mathcal{G}_1, \dots, \mathcal{G}_K$, such as classes or class-balance bins, with $N_g = |\mathcal{G}_g|$. Let S_A be an attribute-driven selected set and let S_H be an attribute-agnostic selected set. Suppose S_A selects $m_{A,g}$ examples from group g and S_H selects $m_{H,g}$ examples from group g. If, conditional on the group counts and on S_A, the attribute-agnostic selector is exchangeable with respect to membership in S_A inside each group, then*

$$\mathbb{E}\big[|S_A \cap S_H| \mid S_A\big] = \sum_{g=1}^{K} \frac{m_{A,g} m_{H,g}}{N_g}. \tag{33}$$

In the ungrouped case, if both selectors choose $m = rN$ examples from a pool of size N, then

$$\mathbb{E}\big[|S_A \cap S_H| \mid S_A\big] = r^2 N. \tag{34}$$

Moreover, as $N \to \infty$ with fixed ratio r, the Jaccard overlap

$$J(S_A, S_H) = \frac{|S_A \cap S_H|}{|S_A \cup S_H|} \tag{35}$$

converges in probability to the random-subset scale

$$J_{\text{rand}}(r) = \frac{r}{2 - r}. \tag{36}$$

Proof. Fix a group $\mathcal{G}_g$ and condition on S_A. Since S_H is exchangeable with respect to membership in S_A within the group, selecting $m_{H,g}$ examples from $\mathcal{G}_g$ gives a hypergeometric intersection with mean

$$m_{H,g} \frac{m_{A,g}}{N_g}. \tag{37}$$

Summing over groups gives the first claim. In the ungrouped case this reduces to $m^2/N = r^2 N$. Hypergeometric concentration gives $|S_A \cap S_H|/N \to r^2$ in probability. Since $|S_A \cup S_H| = 2rN - |S_A \cap S_H|$, the continuous mapping theorem gives

$$\frac{|S_A \cap S_H|}{|S_A \cup S_H|} \to \frac{r^2}{2r - r^2} = \frac{r}{2 - r}. \tag{38}$$

□

Proposition 9 gives a useful diagnostic. If an attribute-aware selector and an attribute-agnostic AL selector have Jaccard overlap close to $J_{\text{rand}}(r)$, then the two selected sets are no more similar than two independent budget-matched subsets at that ratio. This is exactly what one expects when the focused attributes carry selection information not present in the AL variables H_i. Conversely, if a

conventional AL method were already aligned with the discovered attribute slice, its overlap with the selected set from AttriSelect would tend to rise above this random-overlap scale.

The ungrouped value $J_{\mathrm{rand}}(r)$ is a useful lower reference for selectors that share a group allocation over imbalanced groups. Suppose two exchangeable selectors both choose B_g examples from group g, with $\sum_g B_g = rN$ and $\sum_g N_g = N$. Proposition 9 gives expected intersection $\sum_g B_g^2/N_g$, and Cauchy–Schwarz gives

$$\sum_g \frac{B_g^2}{N_g} \geq \frac{\left(\sum_g B_g\right)^2}{\sum_g N_g} = r^2 N, \tag{39}$$

with equality only when B_g/N_g is constant across groups. For fixed total budget, the large-pool Jaccard reference is an increasing function of the intersection size, so class-balanced allocations on imbalanced datasets can have a random-overlap scale slightly above the ungrouped limit. This helps explain why the baseline–baseline averages in Table 15 are consistently above $J_{\mathrm{rand}}(r)$ when the baselines share similar class-balancing effects. When two methods use different group allocations, the full grouped expression in Proposition 9 is the appropriate reference.

Proposition 10 (Non-redundant repair utility)**.** *Assume the utility model from the main text,*

$$U(S) = u_0|S| + (u_T - u_0)|S \cap T|, \qquad u_T > u_0, \tag{40}$$

where T is the target failure slice. Suppose AttriSelect satisfies the deterministic rare-slice enrichment condition in Proposition 3, while an attribute-agnostic selector satisfies the conditional-independence condition in Proposition 1, under the same per-class allocations B_c. Then the expected utility advantage of AttriSelect over the attribute-agnostic selector is at least

$$(u_T - u_0) \sum_c \left[\min\{B_c, |T_c|\} - B_c \frac{|T_c|}{N_c} \right]. \tag{41}$$

This lower bound is strict whenever at least one class has $0 < |T_c| < N_c$ and receives a positive budget that does not occupy the whole class, i.e., $0 < B_c < N_c$. If, in addition, the empirical Jaccard overlap between the two selected sets is close to $J_{\mathrm{rand}}(r)$, then this utility advantage cannot be explained by selecting the same source examples as the attribute-agnostic baseline.

Proof. The main-text enrichment result gives at least $\min\{B_c, |T_c|\}$ target examples for AttriSelect in class c. Proposition 1 gives $B_c|T_c|/N_c$ target examples in expectation for an attribute-agnostic selector under the same class allocation. Since both selectors use the same total budget, the $u_0|S|$ term cancels. Multiplying the difference in target-slice support by the marginal utility gap $u_T - u_0$ gives the stated bound. The strictness condition follows because $\min\{B_c, |T_c|\} > B_c|T_c|/N_c$ whenever a nontrivial target slice receives budget without occupying the whole class. The final statement follows from Proposition 9: near-random Jaccard overlap means the two selectors are not choosing substantially the same subset beyond what the sampling ratio itself forces. □

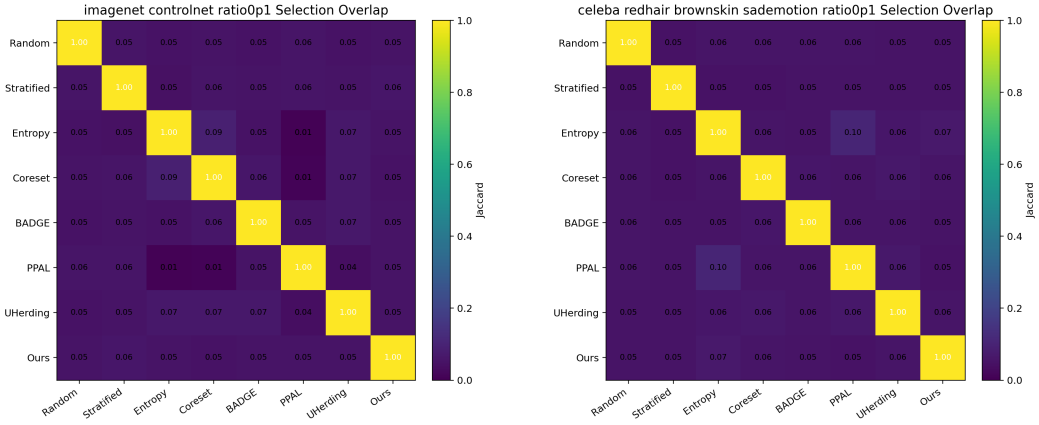
Table 15 summarizes the observed Jaccard overlaps between selected subsets. The t-SNE coordinates are not used in the calculation; the metric is computed only from selected-set membership. Across ratios, the average overlap between AttriSelect and the baselines is close to the random-subset scale $J_{\mathrm{rand}}(r)$ and lower than the average pairwise overlap among baselines. This supports the interpretation that AttriSelect operates along a different semantic information direction rather than imitating a conventional AL heuristic.

Table 15: Selection-overlap diagnostic. "Ours–baseline" is the mean Jaccard overlap between AttriSelect and the non-attribute baselines. "Baseline–baseline" is the mean pairwise Jaccard overlap among non-attribute baselines. Each row averages over ImageNet, CelebA, COCO, and KITTI.

Selection ratio	$J_{\mathrm{rand}}(r)$	Datasets	Ours–baseline	Baseline–baseline
0.10	0.0526	4	0.0533	0.0554
0.30	0.1765	4	0.1766	0.1828
0.50	0.3333	4	0.3323	0.3416

The corresponding heatmaps are shown in Figures 11, 12, and 13. The diagonal entries are one by definition; off-diagonal entries measure pairwise Jaccard overlap between selected source subsets.

These numbers and figures should be read together with the propositions above. They do not prove that lower overlap alone improves a model. Rather, they show that AttriSelect is not redundant with uncertainty, diversity, geometry, detector-difficulty, or gradient-embedding selectors. The improvement mechanism is therefore consistent with the theory: the selector spends budget on an attribute-defined repair signal that the generic AL variables do not explicitly encode.



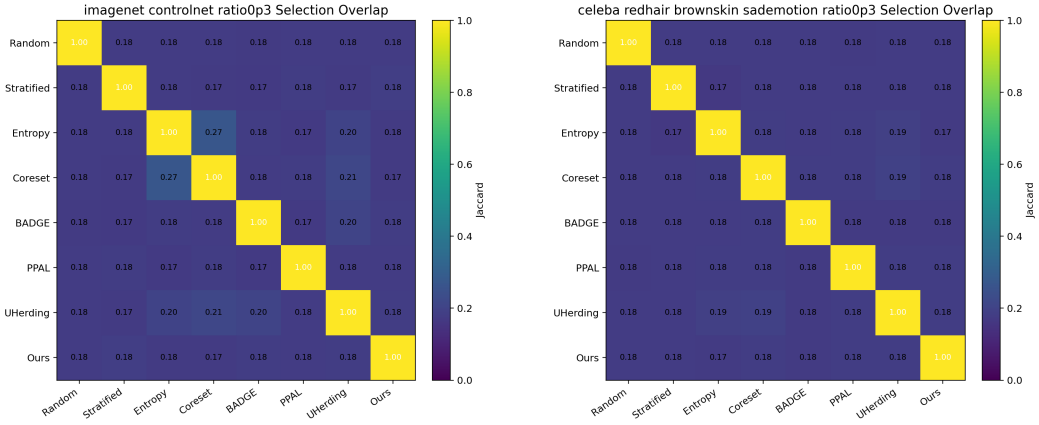
(a) ImageNet

(b) CelebA

(c) COCO

(d) KITTI

Figure 11: Pairwise Jaccard overlap among ratio 0.10 selected subsets.



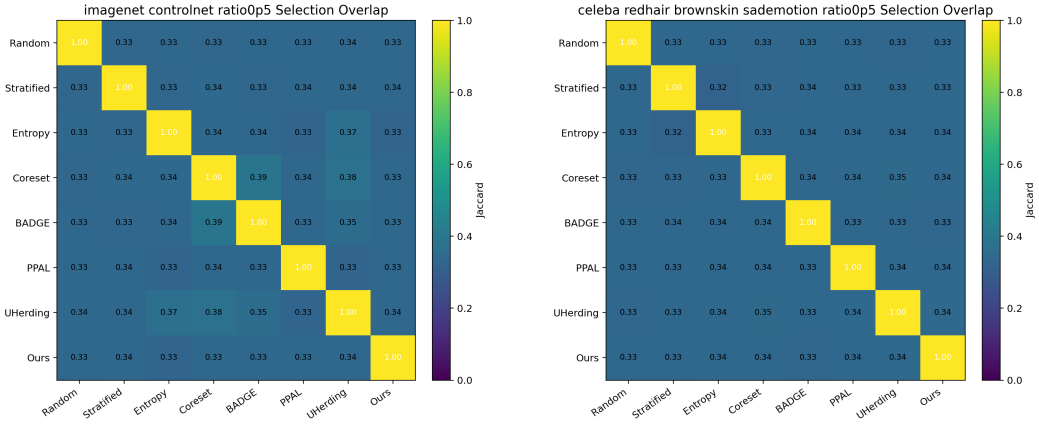
(a) ImageNet

(b) CelebA

(c) COCO

(d) KITTI

Figure 12: Pairwise Jaccard overlap among ratio 0.30 selected subsets.



(a) ImageNet

(b) CelebA

(c) COCO

(d) KITTI

Figure 13: Pairwise Jaccard overlap among ratio 0.50 selected subsets.